Objective Questions

1. **Does any table have missing values or duplicates? If yes how would you handle it ?**

   The data provided possess 0 duplicates although there are missing values in 3 tables.

   Customer table: 47 fax, 29 state and 49 company values are null in the customer table

   Employee table: 1 reports to value is null for employee_id = 1

   Track table: 978 composers are not assigned any value/are null in the track table

   These Null values could be handled easily by using the Coalesce function or UPDATE command in SQL.

```
UPDATE customer SET company = 'Unknown' WHERE company IS NULL;   -- 49 rows affected
UPDATE customer SET state = 'None' WHERE state IS NULL; -- 29 row(s) affected
UPDATE customer SET phone = '+0 000 000 0000' WHERE phone IS NULL; -- 1 row(s) affected
UPDATE customer SET fax = '+0 000 000 0000' WHERE fax IS NULL; -- 47 row(s) affected
UPDATE track SET composer = 'Unknown' WHERE composer IS NULL; -- 978 row(s) affected
```

2. **Find the top-selling tracks and top artist in the USA and identify their most famous genres.**

   Top 10 tracks by units sold in USA

```
select t.name as track_name,ar.name as artist,g.name as genre,sum(il.quantity)
as units_sold
from invoice_line as il join invoice as i on il.invoice_id = i.invoice_id
join track as t on il.track_id = t.track_id
join album as al on t.album_id = al.album_id
join artist as ar on al.artist_id = ar.artist_id
join genre as g on t.genre_id = g.genre_id
join customer as c on i.customer_id = c.customer_id
where c.country = 'usa'
group by t.track_id, t.name, ar.name, g.name
order by units_sold desc
limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

track_name	artist	genre	units_sold
War Pigs	Cake	Alternative	6
You Know I'm No Good (feat. Ghostface Killah)	Amy Winehouse	R&B/Soul	5
Violent Pornography	System Of A Down	Metal	4
End Of The Night	The Doors	Rock	4
Night Of The Long Knives	AC/DC	Rock	4
Scentless Apprentice	Nirvana	Rock	4
Evil Woman	Black Sabbath	Metal	4
I Looked At You	The Doors	Rock	4
Highway Chile	Jimi Hendrix	Rock	4
I Can't Remember	Alice In Chains	Rock	3

Top 5 artists by revenue in USA

```
select ar.name as artist,round(sum(il.unit_price * il.quantity), 2) as revenue_usd
from invoice_line as il
join invoice as i on il.invoice_id = i.invoice_id
join customer as c on i.customer_id = c.customer_id
join track as t on il.track_id = t.track_id
join album as al on t.album_id = al.album_id
join artist as ar on al.artist_id = ar.artist_id
where c.country = 'usa'
group by ar.artist_id, ar.name
order by revenue_usd desc
limit 5;
```

Result Grid | Filter Rows: | Expor

artist	revenue_usd
Van Halen	42.57
R.E.M.	37.62
The Rolling Stones	36.63
Nirvana	34.65
Foo Fighters	33.66

Most famous genre per top USA artist

```
select ar.name as artist, g.name as top_genre,round(sum(il.unit_price *
il.quantity), 2) as genre_revenue
from invoice_line as il
join invoice as i on il.invoice_id = i.invoice_id
join customer as c on i.customer_id = c.customer_id
join track as t on il.track_id = t.track_id
join album as al on t.album_id = al.album_id
join artist as ar on al.artist_id = ar.artist_id
```

```
join genre as g on t.genre_id = g.genre_id
where c.country = 'usa'
group by ar.artist_id, ar.name, g.genre_id, g.name
order by ar.name, genre_revenue desc;
```

Result Grid | Filter Rows: | Export:

artist	top_genre	genre_revenue
AC/DC	Rock	28.71
Accept	Rock	1.98
Aerosmith	Rock	6.93
Aisha Duo	Jazz	0.99
Alanis Morissette	Rock	7.92
Alice In Chains	Rock	20.79
Amy Winehouse	R&B/Soul	15.84
Amy Winehouse	Pop	13.86
Antônio Carlos Jobim	Latin	0.99
Apocalyptica	Metal	7.92
Audioslave	Alternative	6.93

Result 19

Hence, The Top Genre in USA is Rock.

3. **What is the customer demographic breakdown (age, gender, location) of Chinook's customer base?**

```
select country, state, city, COUNT(*) AS customers
from customer
group by country, state, city
order by customers DESC;
```

Result Grid | Filter Rows: | Export:

country	state	city	customers
Czech Republic	None	Prague	2
Brazil	SP	São Paulo	2
USA	CA	Mountain View	2
Germany	None	Berlin	2
France	None	Paris	2
United Kingdom	None	London	2
Canada	QC	Montréal	1
Norway	None	Oslo	1
Austria	None	Vienne	1
Belgium	None	Brussels	1
Denmark	None	Copenhagen	1

Result 20

4. **Calculate the total revenue and number of invoices for each country, state, and city:**

```
select billing_country,billing_state,billing_city,count(invoice_id) as
total_invoices,sum(total) as total_revenue
from invoice
group by billing_country, billing_state, billing_city
```

order by total_revenue desc;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

billing_country	billing_state	billing_city	total_invoices	total_revenue
Czech Republic	None	Prague	30	273.24
USA	CA	Mountain View	20	169.29
United Kingdom	None	London	19	166.32
Germany	None	Berlin	20	158.40
France	None	Paris	18	151.47
Brazil	SP	São Paulo	22	129.69
Ireland	Dublin	Dublin	13	114.84
India	None	Delhi	13	111.87
Brazil	SP	São José dos Campos	13	108.90
Brazil	DF	Brasília	15	106.92
Portugal	None	Lisbon	13	102.96

Result 21

5. Find the top 5 customers by total revenue in each country

```
with cte as (
    select c.customer_id,c.first_name,c.last_name,c.country,sum(i.total) as
revenue,
   dense_rank() over(partition by c.country order by sum(i.total) desc) as rnk
    from customer as c join invoice as i on c.customer_id = i.customer_id
    group by c.customer_id, c.country
)
select * from cte
where rnk <= 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	first_name	last_name	country	revenue	rnk
56	Diego	Gutiérrez	Argentina	39.60	1
55	Mark	Taylor	Australia	81.18	1
7	Astrid	Gruber	Austria	69.30	1
8	Daan	Peeters	Belgium	60.39	1
1	Luís	Gonçalves	Brazil	108.90	1
13	Fernanda	Ramos	Brazil	106.92	2
12	Roberto	Almeida	Brazil	82.17	3
11	Alexandre	Rocha	Brazil	69.30	4
10	Eduardo	Martins	Brazil	60.39	5
3	François	Tremblay	Canada	99.99	1
30	Edward	Francis	Canada	91.08	2

Result 22

6. Identify the top-selling track for each customer

```
with cte as(
    select c.customer_id,t.name as track,sum(il.quantity) as purchases,
   rank() over(partition by c.customer_id order by sum(il.quantity) desc) as rnk
    from customer as c join invoice as i on c.customer_id = i.customer_id
    join invoice_line as il on i.invoice_id = il.invoice_id
    join track as t on il.track_id = t.track_id
    group by c.customer_id, t.name
)
select * from cte
where rnk = 1;
```

Result Grid | Filter Rows: | Export: | Wrap

customer_id	track	purchases	rnk
1	My Generation	1	1
1	Train In Vain	1	1
1	Chuva No Brejo	1	1
1	A Kind Of Magic	1	1
1	So Fast, So Numb	1	1
1	Money	1	1
1	Please Don't Touch	1	1
1	Radio GA GA	1	1
1	A Sua	1	1
1	Straight Out Of Line	1	1
1	October Song	1	1

7. Are there any patterns or trends in customer purchasing behavior (e.g., frequency of purchases, preferred payment methods, average order value)?

a.frequency of purchases

```
select c.customer_id,c.first_name,c.last_name,count(i.invoice_id) as
total_purchases
from customer as c join invoice as i
on c.customer_id = i.customer_id
group by c.customer_id, c.first_name, c.last_name
order by total_purchases desc;
```

Result Grid | Filter Rows: | Export: | W

customer_id	first_name	last_name	total_purchases
5	František	Wichterlová	18
35	Madalena	Sampaio	16
13	Fernanda	Ramos	15
30	Edward	Francis	13
46	Hugh	O'Reilly	13
1	Luís	Gonçalves	13
58	Manoj	Pareek	13
57	Luis	Rojas	13
34	João	Fernandes	13
26	Richard	Cunningham	12
20	Dan	Miller	12

Result 24

b. Average order value

```
select customer_id,round(avg(total), 2) as avg_order_value
from invoice
group by customer_id
order by avg_order_value desc;
```

Result Grid | Filter Rows:

customer_id	avg_order_value
3	11.11
6	10.73
29	10.15
18	9.90
37	9.41
27	9.35
16	9.28
42	9.09
59	8.91
24	8.91
53	8.91

Result 25

8. What is the customer churn rate?

- Customer churn rate measures the percentage of customers who stopped purchasing from the store during a specific period.
- we assume that customers who have not made any purchases in the last 3 months are considered churned customers

```
select count(*) as churned_customers
from customer
where customer_id not in (select distinct customer_id from invoice
   where invoice_date >= date_sub((select max(invoice_date) from
invoice),interval 3 month)
);
```

Result Grid | Filter Rows:

churned_customers
22

9. **Calculate the percentage of total sales contributed by each genre in the USA and identify the best-selling genres and artists.**

Total Sales by Genre in the USA

```
select g.name as genre,sum(il.unit_price * il.quantity) as genre_sales,
round(100 * sum(il.unit_price * il.quantity) /sum(sum(il.unit_price * il.quantity))
over(),2) as percentage_of_total_sales
from invoice as i
join invoice_line as il on i.invoice_id = il.invoice_id
join track as t on il.track_id = t.track_id
join genre as g on t.genre_id = g.genre_id
where i.billing_country = 'usa'
group by g.name
order by genre_sales desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

genre	genre_sales	percentage_of_total_sales
Rock	555.39	53.38
Alternative & Punk	128.70	12.37
Metal	122.76	11.80
R&B/Soul	52.47	5.04
Blues	35.64	3.43
Alternative	34.65	3.33
Latin	21.78	2.09
Pop	21.78	2.09
Hip Hop/Rap	19.80	1.90
Jazz	13.86	1.33
Easy Listening	12.87	1.24

Best-Selling Genre and Artist Together

```
select g.name as genre,ar.name as artist,sum(il.unit_price * il.quantity) as sales
from invoice as i
join invoice_line as il on i.invoice_id = il.invoice_id
join track as t on il.track_id = t.track_id
join genre as g on t.genre_id = g.genre_id
join album as al on t.album_id = al.album_id
join artist as ar on al.artist_id = ar.artist_id
```

```
where i.billing_country = 'usa'
group by g.name, ar.name
order by sales desc;
```

Result Grid | Filter Rows: | Expor

genre	artist	sales
Rock	Van Halen	42.57
Rock	The Rolling Stones	36.63
Rock	Nirvana	34.65
Alternative & Punk	Green Day	31.68
Alternative & Punk	R.E.M.	31.68
Blues	Eric Clapton	30.69
Rock	AC/DC	28.71
Rock	Pearl Jam	28.71
Rock	Jimi Hendrix	27.72
Rock	The Doors	26.73
R&B/Soul	Marvin Gave	25.74

10. Find customers who have purchased tracks from at least 3 different genres

```
select c.customer_id,count(distinct g.genre_id) as genres_purchased
from customer as c
join invoice as i on c.customer_id = i.customer_id
join invoice_line as il on i.invoice_id = il.invoice_id
join track as  t on il.track_id = t.track_id
join genre as g on t.genre_id = g.genre_id
group by c.customer_id
having count(distinct g.genre_id) >= 3;
```

Result Grid | Filter Rows:

customer_id	genres_purchased
1	10
2	14
3	8
4	9
5	13
6	11
7	8
8	10
9	6
10	9
11	8

Result 3

11. Rank genres based on their sales performance in the USA

```
select g.name as genre,sum(il.quantity) as units_sold,round(sum(il.unit_price *
il.quantity), 2) as revenue_usd,
    rank() over (order by sum(il.unit_price * il.quantity) desc) as revenue_rank,
    rank() over (order by sum(il.quantity) desc) as units_rank
from invoice_line as il
join invoice as i on il.invoice_id = i.invoice_id
join customer as c on i.customer_id = c.customer_id
join track as t on il.track_id   = t.track_id
join genre as g on t.genre_id    = g.genre_id
where c.country = 'usa'
group by g.genre_id, g.name
order by revenue_rank;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

genre	units_sold	revenue_usd	revenue_rank	units_rank
Rock	561	555.39	1	1
Alternative & Punk	130	128.70	2	2
Metal	124	122.76	3	3
R&B/Soul	53	52.47	4	4
Blues	36	35.64	5	5
Alternative	35	34.65	6	6
Latin	22	21.78	7	7
Pop	22	21.78	7	7
Hip Hop/Rap	20	19.80	9	9
Jazz	14	13.86	10	10
Easy Listening	13	12.87	11	11

12. Identify customers who have not made a purchase in the last 3 months

```
select c.customer_id,c.first_name,c.last_name,c.email,c.country

from customer as c

where c.customer_id not in(select distinct customer_id from invoice

                where invoice_date >= date_sub((select max(invoice_date) from invoice),interval 3 month)

);
```

Result Grid | Filter Rows: | Edit: | Export/Import:

customer_id	first_name	last_name	email	country
1	Luís	Gonçalves	luisg@embraer.com.br	Brazil
3	François	Tremblay	ftremblay@gmail.com	Canada
4	Bjørn	Hansen	bjorn.hansen@yahoo.no	Norway
7	Astrid	Gruber	astrid.gruber@apple.at	Austria
8	Daan	Peeters	daan_peeters@apple.be	Belgium
9	Kara	Nielsen	kara.nielsen@jubii.dk	Denmark
10	Eduardo	Martins	eduardo@woodstock.com.br	Brazil
11	Alexandre	Rocha	alero@uol.com.br	Brazil
17	Jack	Smith	jacksmith@microsoft.com	USA
18	Michelle	Brooks	michelleb@aol.com	USA
19	Tim	Gover	tgover@apple.com	USA

customer 11

Subjective Questions

1. **Recommend the three albums from the new record label that should be prioritised for advertising and promotion in the USA based on genre sales analysis.**

```
with recommendedalbums as (
   select al.title as album_name,a.name as artist_name,g.name as
genre_name,
         sum(i.total) as total_sales,sum(il.quantity) as total_quantity,
         row_number() over(order by sum(i.total) desc) as sales_rank
   from customer as c
   join invoice as i on c.customer_id = i.customer_id
   join invoice_line as il on i.invoice_id = il.invoice_id
   join track as t on il.track_id = t.track_id
   join album as al on t.album_id = al.album_id
   join artist as a on al.artist_id = a.artist_id
   join genre as g on t.genre_id = g.genre_id
   where c.country = 'usa'
   group by al.title, a.name, g.name
)
select *
from recommendedalbums
order by total_sales desc
limit 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

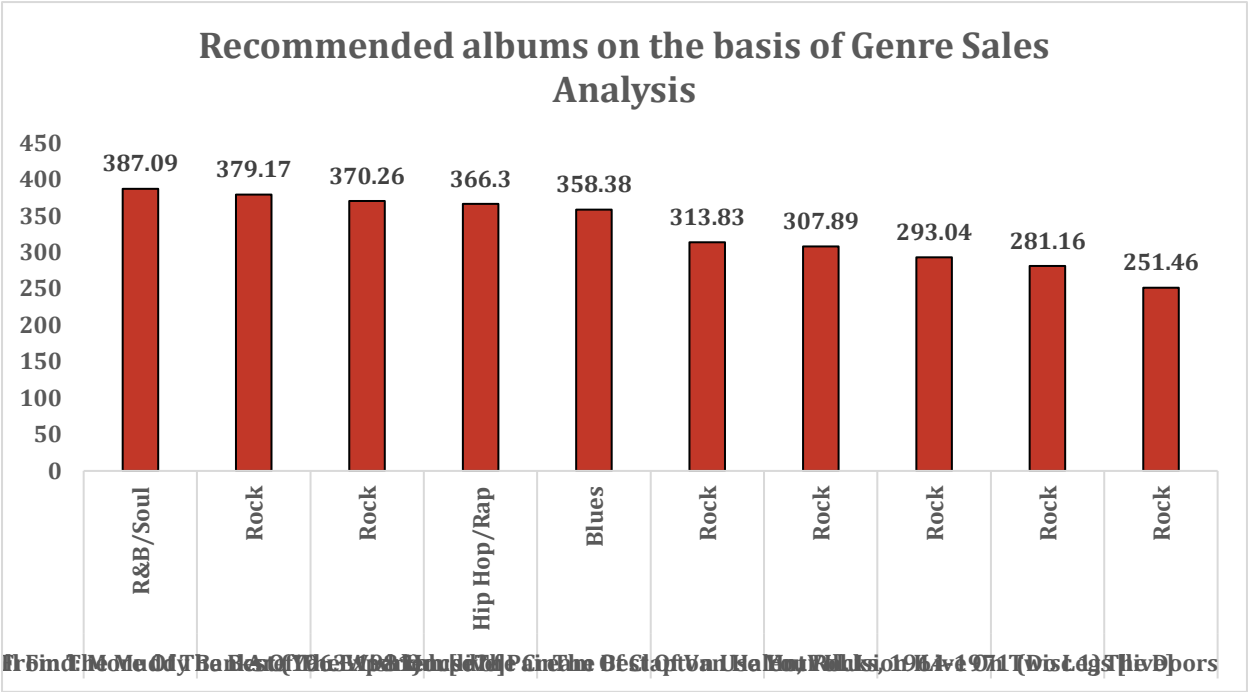
album_name	artist_name	genre_name	total_sales	total_quantity	sales_rank
Seek And Shall Find: More Of The Best (1963-1...	Marvin Gaye	R&B/Soul	387.09	26	1
From The Muddy Banks Of The Wishkah [live]	Nirvana	Rock	379.17	28	2
Are You Experienced?	Jimi Hendrix	Rock	370.26	28	3

Recommendation:

The top 3 albums that should be prioritised for advertisements and promotions in the USA based on genre analysis are:-

1. **Seek And Shall Find: More of the Best** (By Marvin Gayne)
2. **From The Muddy Banks Of The Wishkah** (By Nirvana)
3. **Are You Experienced?** (By Jimi Hendrix)
   These albums are from the genre R&B/Soul and ROCK respectively as it is the most popular genre in the USA

Visualization :



2. **Determine the top-selling genres in countries other than the USA and identify any commonalities or differences**.

- in usa

```
select i.billing_country,g.name as genre,sum(il.unit_price * il.quantity) as revenue
from invoice as i
join invoice_line as il on i.invoice_id = il.invoice_id
join track as t on il.track_id = t.track_id
join genre as g on t.genre_id = g.genre_id
where i.billing_country = 'usa'
group by i.billing_country, g.name
order by revenue desc
limit 10;
```

-- other than usa

```
select i.billing_country,g.name as genre,sum(il.unit_price * il.quantity) as revenue
from invoice as i
join invoice_line as il on i.invoice_id = il.invoice_id
join track as t on il.track_id = t.track_id
join genre as g on t.genre_id = g.genre_id
where i.billing_country <> 'usa'
group by i.billing_country, g.name
order by revenue desc
limit 10;
```

Result Grid | Filter Rows: | Exp

billing_country	genre	revenue
USA	Rock	555.39
USA	Alternative & Punk	128.70
USA	Metal	122.76
USA	R&B/Soul	52.47
USA	Blues	35.64
USA	Alternative	34.65
USA	Pop	21.78
USA	Latin	21.78
USA	Hip Hop/Rap	19.80
USA	Jazz	13.86

Result Grid | Filter Rows: | Export:

billing_country	genre	revenue
Canada	Rock	329.67
France	Rock	208.89
Brazil	Rock	202.95
Germany	Rock	192.06
United Kingdom	Rock	164.34
Czech Republic	Rock	141.57
Portugal	Rock	106.92
India	Rock	100.98
Brazil	Alternative & Punk	73.26
Brazil	Metal	72.27

Commonalities:

- Rock dominates in almost all the countries listed, indicating a strong global preference for Rock music outside the USA.
- Metal and Alternative & Punk are also popular in Brazil, showing a more diverse taste in music genres in that country.
- Rock is the top genre in every country mentioned except for the specific instances where Metal or Alternative & Punk are highlighted.

Differences:

- Brazil stands out with significant sales in both Alternative & Punk and Metal genres, in addition to Rock, showing a more varied musical taste compared to other countries.
- Metal has a notable presence in Canada and Brazil but does not appear as a top genre in other countries listed.
- The sales volume for genres other than Rock is generally lower in all non-USA countries, indicating Rock's dominance.

3. **Customer Purchasing Behavior Analysis: How do the purchasing habits (frequency, basket size, spending amount) of long-term customers differ from those of new customers? What insights can these patterns provide about customer loyalty and retention strategies?**

```
WITH cte as
(
SELECT i.customer_id, MAX(invoice_date), MIN(invoice_date),
abs(TIMESTAMPDIFF(MONTH, MAX(invoice_date), MIN(invoice_date)))
time_for_each_customer, SUM(total) sales, SUM(quantity) items,
COUNT(invoice_date) frequency FROM invoice i
LEFT JOIN customer c on c.customer_id = i.customer_id
LEFT JOIN invoice_line il on il.invoice_id = i.invoice_id
GROUP BY 1
ORDER BY time_for_each_customer DESC
),
average_time as
(
SELECT AVG(time_for_each_customer) average FROM cte
),-- 1244.3220 Days OR 40.36 Months
categorization as
(
SELECT *,
CASE
WHEN time_for_each_customer > (SELECT average from average_time)
THEN "Long-term Customer" ELSE "Short-term Customer"
END category
FROM cte
)
SELECT category, SUM(sales) total_spending, SUM(items) basket_size,
COUNT(frequency) frequency FROM categorization
GROUP BY 1;
```

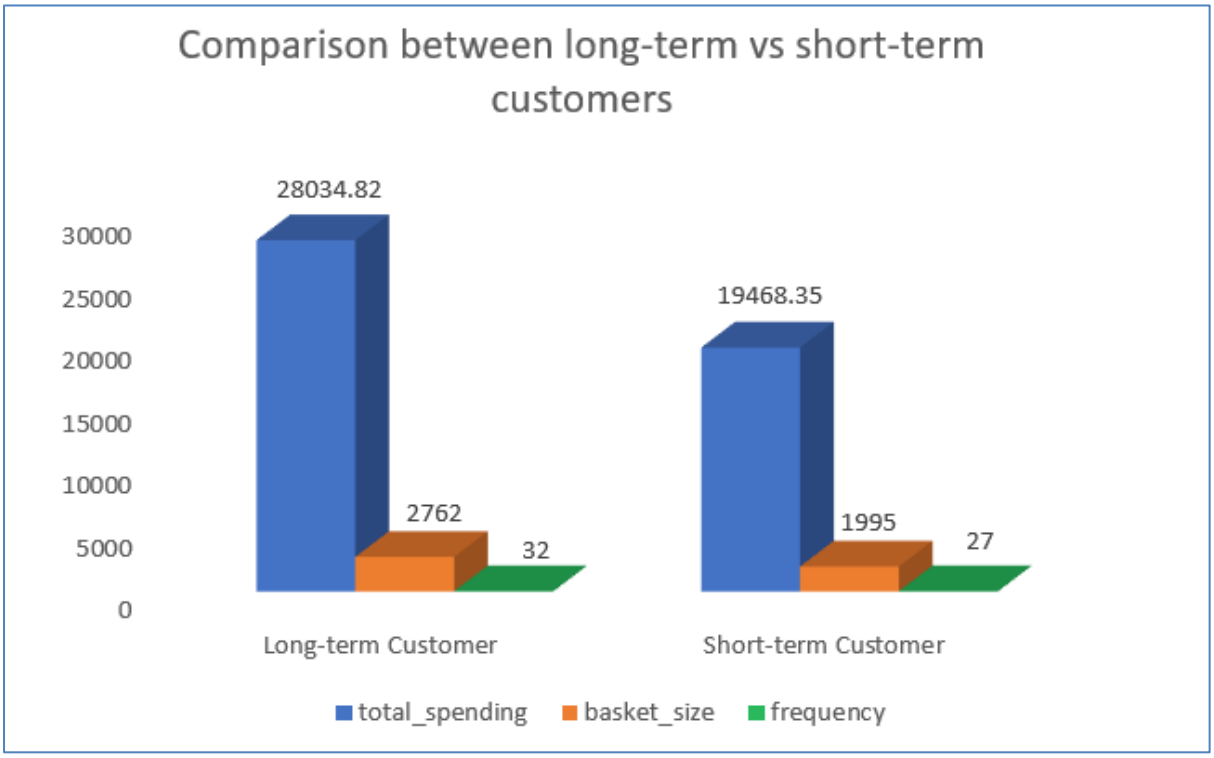
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_spending	basket_size	frequency
Long-term Customer	28034.82	2762	32
Short-term Customer	19468.35	1995	27

Insights:

- **Purchase Frequency**: By comparing avg_purchase_frequency, we can determine whether long-term customers make purchases more frequently than new customers.
- **Basket Size**: The avg_basket_size indicates whether long-term customers tend to buy more items per order compared to new customers

- **Spending Amount**: The avg_spending_amount helps in understanding whether long-term customers spend more money overall, which could be a sign of loyalty and satisfaction.
- **Order Value**: The avg_order_value reveals whether long-term customers place higher-value orders, which could indicate trust and a preference for the product offerings.



Recommendation:

These insights can inform customer loyalty and retention strategies, such as offering personalized promotions to long-term customers or targeted marketing campaigns to encourage repeat purchases from new customers.

4. Product Affinity Analysis: Which music genres, artists, or albums are frequently purchased together by customers? How can this information guide product recommendations and cross-selling initiatives?

```
select g1.name as genre_1,g2.name as genre_2,count(distinct i.customer_id)
as shared_customers
from invoice_line as il1
join invoice as i on il1.invoice_id = i.invoice_id
join track as t1 on il1.track_id = t1.track_id
join genre g1 on t1.genre_id = g1.genre_id
join invoice_line as il2 on il2.invoice_id = i.invoice_id and il2.track_id <>
il1.track_id
join track as t2 on il2.track_id = t2.track_id
join genre as g2 on t2.genre_id = g2.genre_id and g2.genre_id > g1.genre_id
```

```
group by g1.name, g2.name
order by shared_customers desc
limit 15;
```

Result Grid | Filter Rows: | Export:

genre_1	genre_2	shared_customers
Rock	Metal	59
Rock	Alternative & Punk	57
Metal	Alternative & Punk	55
Rock	Latin	49
Rock	Alternative	47
Metal	Latin	46
Rock	R&B/Soul	42
Rock	Jazz	40
Metal	R&B/Soul	38
Metal	Alternative	38
Alterna...	Latin	37

Result 18

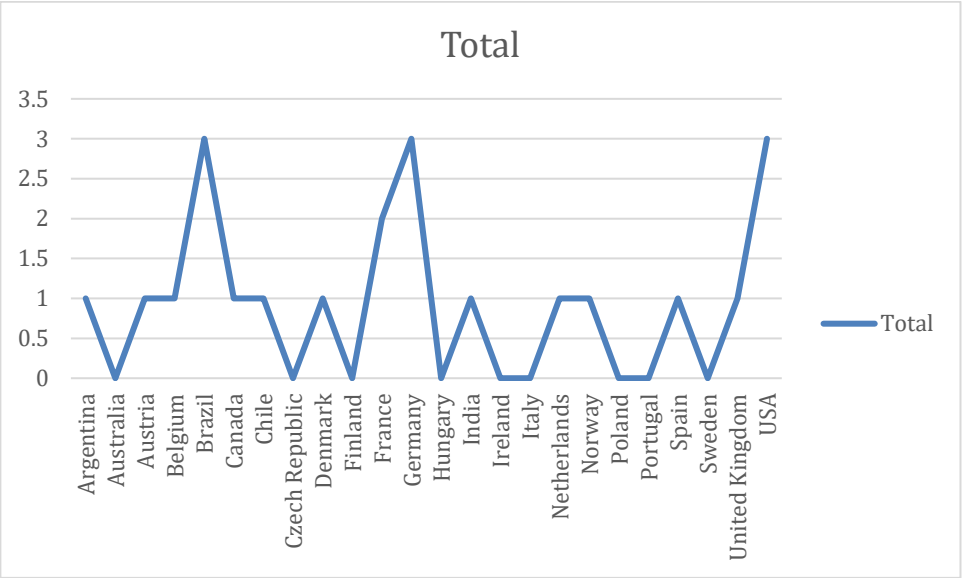
5. **Regional Market Analysis: Do customer purchasing behaviors and churn rates vary across different geographic regions or store locations? How might these correlate with local demographic or economic factors?**

```
select c.country,count(distinct c.customer_id) as total_customers,
count(distinct case when i.invoice_date >= date_sub((select max(invoice_date)
from invoice), interval 3 month) then i.customer_id
   end) as active_customers,
count(distinct c.customer_id) -count(distinct case when i.invoice_date >=
date_sub((select max(invoice_date) from invoice), interval 3 month)
       then i.customer_id
   end) as churned_customers,
count(i.invoice_id) as total_orders,sum(i.total) as total_revenue,round(avg(i.total),
2) as avg_order_value
from customer as c
left join invoice as i
on c.customer_id = i.customer_id
group by c.country
order by total_revenue desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	total_customers	active_customers	churned_customers	total_orders	total_revenue	avg_order_value
USA	13	10	3	131	1040.49	7.94
Canada	8	7	1	76	535.59	7.05
Brazil	5	2	3	61	427.68	7.01
France	5	3	2	50	389.07	7.78
Germany	4	1	3	41	334.62	8.16
Czech Republic	2	2	0	30	273.24	9.11
United Kingdom	3	2	1	28	245.52	8.77
Portugal	2	2	0	29	185.13	6.38
India	2	1	1	21	183.15	8.72
Ireland	1	1	0	13	114.84	8.83
Spain	1	0	1	11	98.01	8.91

Result 20



6. **Customer Risk Profiling: Based on customer profiles (age, gender, location, purchase history), which customer segments are more likely to churn or pose a higher risk of reduced spending? What factors contribute to this risk?**

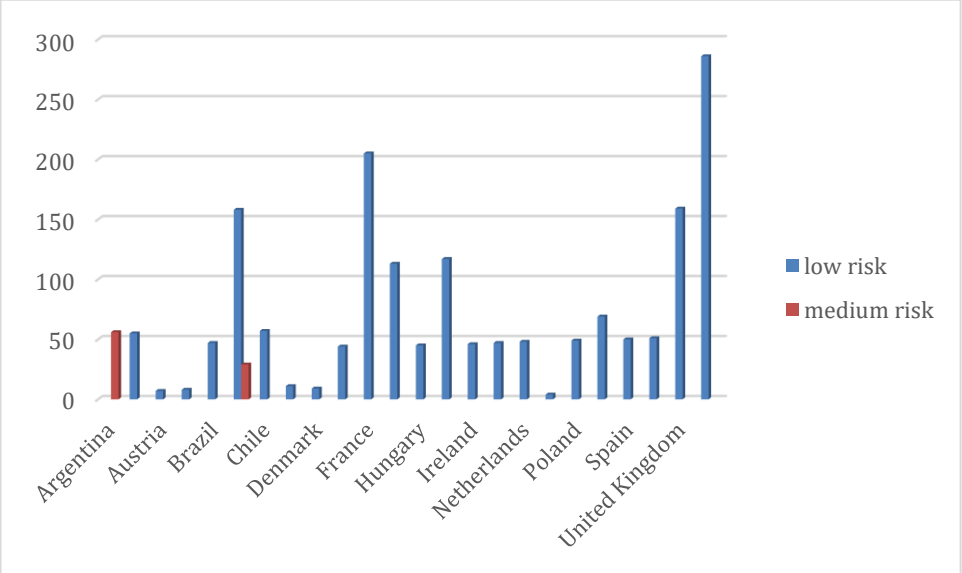
```
with customer_metrics as(
   select c.customer_id,c.first_name,c.last_name,c.country,count(i.invoice_id)
as purchase_frequency,
         sum(i.total) as total_spent,max(i.invoice_date) as last_purchase_date
   from customer as c
   left join invoice as i
   on c.customer_id = i.customer_id
   group by c.customer_id, c.first_name, c.last_name, c.country
)
select
customer_id,first_name,last_name,country,purchase_frequency,total_spent,la
st_purchase_date,
case when purchase_frequency <= 2 and last_purchase_date <=
date_sub((select max(invoice_date) from invoice), interval 3 month)
      then 'high risk'
```

```
when purchase_frequency <= 5 then 'medium risk'
else 'low risk'
end as risk_segment
from customer_metrics
order by risk_segment;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	first_name	last_name	country	purchase_frequency	total_spent	last_purchase_date	risk_segment
1	Luís	Gonçalves	Brazil	13	108.90	2020-07-24 00:00:00	low risk
2	Leonie	Köhler	Germany	11	82.17	2020-11-21 00:00:00	low risk
3	François	Tremblay	Canada	9	99.99	2020-05-16 00:00:00	low risk
4	Bjørn	Hansen	Norway	9	72.27	2020-02-04 00:00:00	low risk
5	František	Wichterlová	Czech Republic	18	144.54	2020-11-12 00:00:00	low risk
6	Helena	Holý	Czech Republic	12	128.70	2020-10-23 00:00:00	low risk
7	Astrid	Gruber	Austria	9	69.30	2020-08-26 00:00:00	low risk
8	Daan	Peeters	Belgium	7	60.39	2019-09-21 00:00:00	low risk
9	Kara	Nielsen	Denmark	10	37.62	2020-01-29 00:00:00	low risk
10	Eduardo	Martins	Brazil	12	60.39	2020-06-25 00:00:00	low risk
11	Alexandre	Rocha	Brazil	10	69.30	2020-06-24 00:00:00	low risk

Result 21



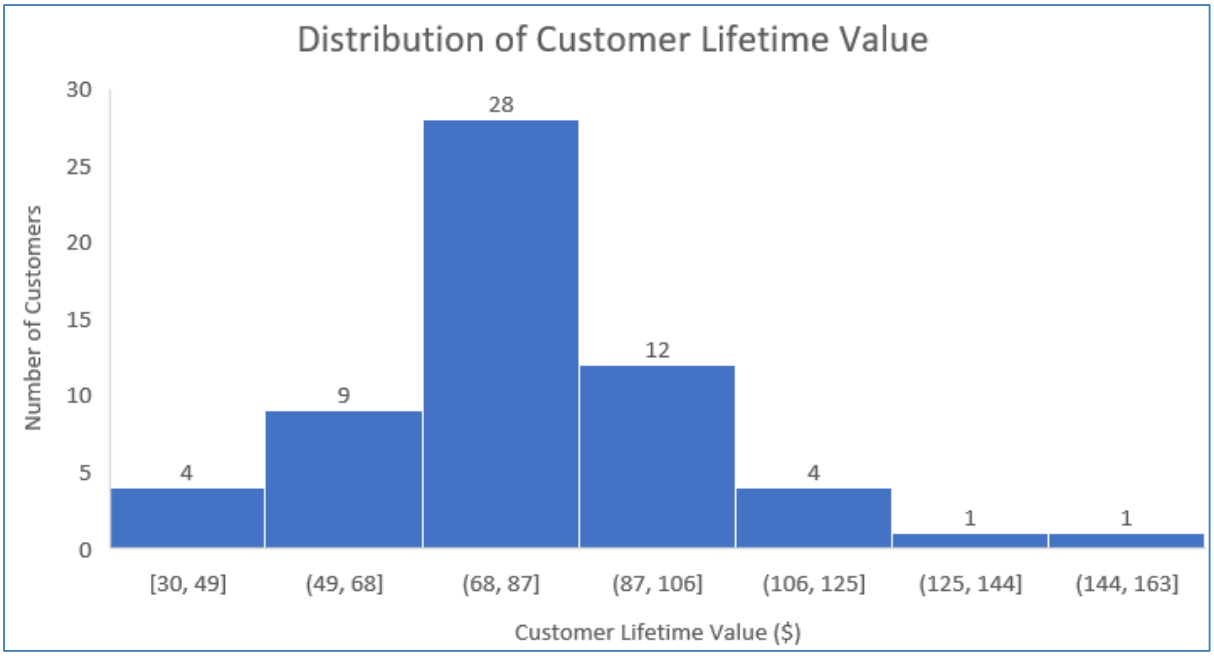
7. Customer Lifetime Value Modeling: How can you leverage customer data (tenure, purchase history, engagement) to predict the lifetime value of different customer segments? This could inform targeted marketing and loyalty program strategies. Can you observe any common characteristics or purchase patterns among customers who have stopped purchasing?

```
with customer_profile as (
select c.customer_id,concat(c.first_name, ' ', c.last_name) as
customers,c.country,coalesce(c.state,'n.a') as state,
c.city,min(i.invoice_date) as first_purchase_date,max(i.invoice_date) as
last_purchase_date,
datediff(max(i.invoice_date), min(i.invoice_date)) as
customer_tenure_days,count(i.invoice_id) as total_purchases,
sum(i.total) as total_spending,round(avg(i.total),2) as avg_order_value
```

```
from customer c left join invoice i on c.customer_id = i.customer_id
group by c.customer_id
),
customer_lifetime_value as (
   select
cp.customer_id,cp.customers,cp.country,cp.state,cp.city,cp.customer_tenure_
days,cp.total_purchases,cp.total_spending,cp.avg_order_value,
         case when cp.customer_tenure_days >= 365 then 'long-term'else
'short-term'
       end as customer_segment,
         case when cp.last_purchase_date < date_sub(curdate(), interval 1
year) then 'churned'else 'active'
         end as customer_status,
         round((cp.total_spending / greatest(cp.customer_tenure_days, 1)) *
365,2) as predicted_annual_value,cp.total_spending as lifetime_value
   from customer_profile cp
),
segment_analysis as (
   select customer_segment,customer_status,count(customer_id) as
num_customers,avg(customer_tenure_days) as
avg_tenure_days,avg(total_spending) as avg_lifetime_value,
   avg(predicted_annual_value) as avg_predicted_annual_value
   from customer_lifetime_value
   group by customer_segment, customer_status
),
churn_analysis as (
   select country,state,city,customer_segment,count(customer_id) as
churned_customers,avg(total_spending) as avg_lifetime_value
   from customer_lifetime_value
   where customer_status = 'churned'
   group by country, state, city, customer_segment
)
select *
from customer_lifetime_value
order by lifetime_value desc;
```

customer_id	customers	country	state	city	customer_tenure_days	total_purchases	total_spending	avg_order_value	customer_segment	customer_status	predicted_annual_value	lifetime_value
5	František Wichterlová	Czech Republic	None	Prague	1263	18	144.54	8.03	long-term	churned	41.77	144.54
6	Helena Holý	Czech Republic	None	Prague	1149	12	128.70	10.73	long-term	churned	40.88	128.70
46	Hugh O'Reilly	Ireland	Dublin	Dublin	1393	13	114.84	8.83	long-term	churned	30.09	114.84
58	Manoj Pareek	India	None	Delhi	1240	13	111.87	8.61	long-term	churned	32.93	111.87
1	Luís Gonçalves	Brazil	SP	São José dos Campos	1275	13	108.90	8.38	long-term	churned	31.18	108.90
13	Fernanda Ramos	Brazil	DF	Brasília	1290	15	106.92	7.13	long-term	churned	30.25	106.92
34	João Fernandes	Portugal	None	Lisbon	1326	13	102.96	7.92	long-term	churned	28.34	102.96
3	François Tremblay	Canada	QC	Montréal	1180	9	99.99	11.11	long-term	churned	30.93	99.99
42	Wyatt Girard	France	None	Bordeaux	1406	11	99.99	9.09	long-term	churned	25.96	99.99
53	Phil Hughes	United Kingdom	None	London	1420	11	98.01	8.91	long-term	churned	25.19	98.01
50	Enrique Muñoz	Spain	None	Madrid	1181	11	98.01	8.91	long-term	churned	30.29	98.01

Result 22



Insights:

1. **Customer Segments and Predicted Lifetime Value:**
- High Lifetime Value Customers: Example: František Wichterlová from Prague. High spending and average order value. Need targeted retention strategies.
2. Low Lifetime Value Customers: Example: Mark Philips from Edmonton. Lower spending and order values. More price-sensitive or less engaged.
3. **Common Characteristics of Churned Customers:**
- Long-Term Customers: Churn can occur even with long tenure. Example: Helena Holý from Prague.
- Moderate Spending: Customers with moderate spending may churn due to better alternatives. Example: Camille Bernard from Paris.
- High Risk, Low Engagement: High predicted lifetime value but low engagement. Example: Robert Brown from Toronto.
4. **Observed Purchase Patterns:**
- Higher Engagement: Frequent purchases reduce churn risk.
- Lower Order Value: Customers with low average order values but long tenure may be at risk if they don't perceive value.
5. **Targeted Marketing and Loyalty Strategies:**
- Personalized Offers: For high CLV customers, use targeted promotions.
- Loyalty Programs: Reward moderate spenders to increase engagement.
- Reactivation Campaigns: For long-term, low-spending customers, offer incentives to boost spending.

8. If data on promotional campaigns (discounts, events, email marketing) is available, how could you measure their impact on customer acquisition, retention, and overall sales?

If data on promotional campaigns were available, I would analyze their impact on:

1. **Customer Acquisition**:

- Tracking the increase in customer count over time.
- Monitoring participation in events across different locations.
- Evaluating the rise in click-through rates from email marketing campaigns.

2. **Customer Retention**:
   - Identifying the number of returning customers who attended events and purchased tracks after a long gap.
   - Assessing the impact of discounts on retaining existing customers.
3. **Sales**:
   - Measuring the sales growth attributed to promotional campaigns.
   - Analyzing trends to identify the most effective campaigns.
   - Tracking new customer acquisition driven by discounts.

This would provide valuable insights into the effectiveness of promotional strategies.

9. How would you approach this problem, if the objective and subjective questions weren't given?

If no predefined questions are given, the approach to solving the problem involves a systematic, exploratory analysis of available data:

1. **Understand Data Schema**
   - Identify entities and relationships (e.g., Employee, Customer, Invoice).
   - Determine key metrics like sales performance and customer demographics.
2. **Data Exploration and Quality Assessment**
   - Check for missing values and duplicates.
   - Clean data by imputing or removing incomplete records.
3. **Sales Performance Analysis**
   - Analyze total sales, average order value, and top-selling tracks across time, geography, and customer segments.
4. **Customer Analysis**
   - Analyze demographics and segment customers by purchase history.
   - Calculate churn rates and model Customer Lifetime Value (CLV).
5. **Market and Regional Analysis**
   - Assess sales performance across regions and correlate with external factors like demographics.
   - Product Affinity and Cross-Selling
   - Identify frequently purchased product combinations for cross-selling opportunities.
6. **Promotional Impact Analysis**
   - Measure the impact of promotions on sales and customer behavior.
7. **Strategic Insights and Recommendations**
   - Identify growth opportunities, optimize pricing, and develop targeted marketing strategies.

8. **Documentation and Reporting**
    - Summarize findings with visualizations and provide actionable recommendations.

Conclusion

This approach involves comprehensive data exploration to uncover insights that guide strategic decisions, enhance customer retention, optimize sales, and identify growth opportunities. The process is iterative and evolves with new insights.

10. How can you alter the "Albums" table to add a new column named "ReleaseYear" of type INTEGER to store the release year of each album?

We can use the ALTER TABLE command to add a new column to the Album table.

```
ALTER TABLE album
ADD COLUMN ReleaseYear INT(4);

SELECT * FROM album;
```

album_id	title	artist_id	ReleaseYear
1	For Those About To Rock We Salute You	1	NULL
2	Balls to the Wall	2	NULL
3	Restless and Wild	2	NULL
4	Let There Be Rock	1	NULL
5	Big Ones	3	NULL
6	Jagged Little Pill	4	NULL
7	Facelift	5	NULL
8	Warner 25 Anos	6	NULL
9	Plays Metallica By Four Cellos	7	NULL
10	Audioslave	8	NULL
11	Out Of Exile	8	NULL
12	BackBeat Soundtrack	9	NULL
13	The Best Of Billy Cobham	10	NULL
14	Alcohol Fueled Brewtality Live! [Disc 1]	11	NULL
15	Alcohol Fueled Brewtality Live! [Disc 2]	11	NULL
16	Black Sabbath	12	NULL
17	Black Sabbath Vol. 4 (Remaster)	12	NULL
18	Body Count	13	NULL

11. Chinook is interested in understanding the purchasing behavior of customers based on their geographical location. They want to know the average total amount spent by customers from each country, along with the number of customers and the average number of tracks purchased per customer. Write an SQL query to provide this information.

select c.country,count(distinct c.customer_id) as

```
total_customers,round(avg(customer_spending.total_spent),2) as
avg_amount_spent_per_customer,round(avg(customer_spending.total_tracks
),2) as avg_tracks_per_customer

from customer as c join

(

    select i.customer_id,sum(i.total) as total_spent,count(il.track_id) as
total_tracks

    from invoice as i

    join invoice_line as il on i.invoice_id = il.invoice_id

    group by i.customer_id

) customer_spending

on c.customer_id = customer_spending.customer_id

group by c.country

order by avg_amount_spent_per_customer desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	total_customers	avg_amount_spent_per_customer	avg_tracks_per_customer
Czech Republic	2	1591.92	138.00
Ireland	1	1433.52	116.00
Spain	1	1076.13	99.00
India	2	943.97	92.50
Australia	1	940.50	82.00
Chile	1	912.78	98.00
Portugal	2	877.64	93.50
Germany	4	860.31	84.50
United Kingdom	3	832.92	82.67
Hungary	1	830.61	79.00
Brazil	5	811.80	86.40